WEEK 3 · DAY 1 · INSTRUCTOR NOTES

Alarms That Cry Wolf

For the trainer only.

The one idea for today: *a fixed alarm limit is already a model, and a model that ignores context cries wolf.* The fix is to add context (which state is the satellite in?), not to widen the limit.

At a glance

Length	~2 hours
Participants leave able to	build an alarm as a True/False column, count and time its alarms, add context with `&`, derive a limit from data (mean − 3σ), and name the three anomaly shapes
Covers (original)	Week 4 Code Book sections **1. Bus power** and **4. RF link quality**; Week 4 quiz topic *what your alarm software already does* (questions 1–6 of `week-04-mcq.csv`)
Datasets	`week-04/data/bus_power_telemetry_hourly.csv` (identical to the week-03 copy), `rf_link_quality_hourly.csv`, plus a peek at `twta_thermal_daily.csv` and `battery_health_daily.csv` (week-04 versions, which contain the faults)
Withhold	The label file `telemetry_anomaly_labels.csv`. It's the answer key, introduced on **W4D1**. Today's participant materials never show it

Run of show (2 hours)

Time	Block	What happens	Slides
0:00	**Hook**	NigComSat-1 lost in 2008 to a solar-array anomaly → "watch the array"	1–3
0:10	**The alarm you already have**	A limit is a model; the four tricks of alarm software; "where did that number come from?"	4–7
0:22	**Cry wolf**	Eclipse = the satellite's NEPA; 50 alarms, 46 useless; alarm fatigue; the context fix; battery temperature	8–13
0:35	**Where should the limit be?**	3σ from sunlit data; what IS normal?; rain vs outage; persistence trap; leaking the answer; three shapes	14–20
0:50	**Hands-on**	Part A (20 min), Part B (15 min), Part C (25 min)	21
1:50	**Wrap-up**	Recap; ask each person to name one alarm in their unit that cries wolf	22

Explain it like this (in plain English)

- **A threshold is a model.** "Below 20 A = bad" is a tiny anomaly detector. It can be good or bad, and we can measure which.
- **Eclipse.** Twice a year, around the equinoxes, the Earth's shadow crosses the satellite once a night (here at 21:00 UTC, 22:00 in Lagos). The array current drops to zero and the battery takes over. It's the satellite's "NEPA has taken light", and it is completely normal.
- **Why the floor cries wolf.** A 20 A floor fires every eclipse night (46 times) and on the 4 real dropout hours. 46 of 50 alarms are useless, so the NOC learns to ignore the channel. That's alarm fatigue.
- **The fix.** `& (bus["eclipse_flag"] == 0)` : only check the limit in sunlight. 50 → 4 alarms, all real. Don't widen the limit; that would hide the dropouts too.
- **Where should the limit be?** Use normal data: mean − 3σ. But "normal" is slippery: the 4 dropouts alone stretch σ from about 0.17 to about 0.52 A, and the post-eclipse recharge boost stretches it to 0.59. Define normal per state. Day 2 does this properly with control charts.
- **Rain is not a fault.** In heavy rain C/N falls whether or not the link drops. Link margin below 0 dB is the honest outage definition.
- **Persistence.** "Only alarm after 2 hours" sounds wise, but every August outage lasted exactly 1 hour, so it catches none of them.
- **Leaking the answer.** Uplink power rises *because* of rain (power control). A detector fed with it learns "is it raining?", not "is the link broken?".
- **Three shapes.** Spike (a short dip), level shift (a new normal), drift (a slow slide). Fixed limits only really catch spikes.

Numbers you'll quote (all computed from the data)

What	Value
Bus power file	2,880 hourly rows, 1 Aug – 28 Nov 2025, no gaps
Eclipse hours	**46**, all at 21:00 UTC, from 1 Sep to 16 Oct
Floor `array < 20 A`	**50** alarms = **46** eclipse + **4** real
With context (`eclipse_flag == 0`)	**4** alarms, all real
SoC floor `< 82%`	**18** alarms, **all 18** in eclipse (SoC ranges 77.3–99.6% across eclipse hours)
Battery temp `> 22 °C`	**50** alarms = **46** in the hour after eclipse + **4** real
Sunlit array current	mean **28.21 A**, σ **0.594 A** → 3σ band **26.43–29.99 A** → **4** hours outside (the dropouts)
Same, without recharge hours and dropouts	σ **0.169 A** → band **27.65–28.66 A** → **143** sunlit alarms, **138** of them recharge hours
RF, August	744 hours, **56** rainy, **12** outages (margin < 0 dB), every outage 1 hour long, every one in rain ≥ 39.3 mm/hr

What	Value
3σ on C/N, all hours	floor **10.76 dB** → **18** alarms, **12** real, 6 just rain
3σ on C/N, dry-hour baseline	floor **14.06 dB** → **49** alarms, **12** real (**76%** false)
2-hours-in-a-row rule	catches **0** of 12 outages
Uplink power vs rain	correlation **0.97**; mean **41.2** dBm dry vs **44.1** dBm in rain
TWTA (week-04)	3σ ceiling **69.8 °C**, hottest day **68.2 °C** (3 Sep 2024): fires **0** times; level shift **65.2** → **67.4 °C** in May 2025
Battery capacity (week-04)	**103.3** → **95.4 Ah** (min 95.39); a 95 Ah limit fires **0** times

🔒 The hidden events (instructor only)

These are in the label file that participants see on W4D1. Today they find them without it.

Event	When	What	How it shows today
EV01	18 Aug 2025 09:00	Solar-array dropout (minor)	array 15.34 A, battery 23.0 °C
EV02	28 Sep 2025 14:00–15:00	Two-hour dropout with thermal excursion (major)	array ~15.5 A, battery ~22.9 °C (the bonus date)
EV03	31 Oct 2025 03:00	Solar-array dropout (minor)	array 15.54 A
EV04	3–4 Sep 2024	TWTA transient spike	68.15 °C, below the 3σ ceiling
EV05	from 12 May 2025	TWTA level shift (cooling-loop blockage)	the "level shift" chart in C4
EV06	from 1 Apr 2025	Battery fade rate roughly doubles	the "drift" chart in C4

Answer key

Task	Answer	Common mistake → what to say
A1	`bus["eclipse_flag"].sum()` → **46**	–
A2	`floor_alarm.sum()` → **50**	–
A3	`(floor_alarm & (bus["eclipse_flag"] == 1)).sum()` → **46**	`and` instead of `&` → "ambiguous truth value" error
A4	`(bus["battery_soc_pct"] < 82).sum()` → **18**	–
A5	`SUNLIT = 0` → 4 alarms left	Writing `"0"` (text)
B1	`sunlit.mean() - 3 * sunlit.std()` → **26.43**	Using `bus["solar_array_current_a"]` (includes eclipse: band 17.0–38.5 A)

Task	Answer	Common mistake → what to say
B2	`(sunlit < low_limit).sum()` → **4**	–
B3	`(hot & after_eclipse).sum()` → **46**	–
C1	`(cn_flags.sum(), (cn_flags & outage).sum())` → **(18, 12)**	Giving one number
C2	`two_in_a_row.sum()` → **0**	They expect a positive number. That's the lesson
C3	`"no"`	–
C4	`("spike", "level shift", "drift")`	Mixing up shift and drift. A shift jumps, a drift slides
C5	`(bat["est_capacity_ah"] < 95).sum()` → **0**	–
🌶️	`"2025-09-28"`	–

Quiz link

Week 4 quiz (`Time-series/MCQ/week-04-mcq.csv`), topic **"what your alarm software already does"**, questions 1–6: threshold is a model (slide 5), seasonal false alarms (slide 10), silent degradation (slide 20 + C5), three shapes (slide 20 + C4), leaking the answer (slide 19 + C3), where thresholds come from (slide 7). Quiz 2 runs on **W4D1**.

Troubleshooting

Problem	Fix
"The truth value of a Series is ambiguous"	They used `and` / `or`. Use `&` / `|` with brackets around each comparison
A5 gives 0 alarms	They wrote `SUNLIT = 1` or `"0"`. Sunlit is the number `0`
The C4 charts look empty	`tw` / `bat` didn't load. Re-run the C4 code cell
Someone asks "can we see the real answer key?"	"On Week 4 Day 1. First you build detectors without peeking, just like real life"

Notes on the original material

- **The SoC example doesn't match the data.** The Week 4 code book and quiz say a SoC < 82% floor "fires on all 46 eclipse hours". It fires **18** times (all in eclipse). Eclipse SoC ranges 77.3–99.6%; only a floor above 99.6% would catch all 46. The lesson (correct but useless) holds, so slides use the honest 18 and the array-current floor for the 46.
- **The sunlit standard deviation.** The code book says σ ≈ 0.06 A and a band of 27.6–28.0 A. The data gives σ = **0.594 A** and **26.43–29.99 A**, because the post-eclipse recharge hours (up to 29.8 A) and the dropouts themselves are in "sunlit". Even with both removed, σ is 0.169 A. Monthly σ is 0.07–0.10 A, closest to the original's figure.

- **RF naive rule.** The code book says a 3σ rule on C/N flags "roughly all 56 rainy hours, 79% false alarms". With σ from all hours it flags **18** (6 false). The original figure matches a *dry-hour* baseline: **49** flags, 76% false. Both are shown on slide 17.